

Final Project Proposal

Title: Climate Change and U.S. Billion-Dollar Disasters: Trend Estimation, Double Bootstrap Inference, Neural Networks, and FDR Analysis in R

Overview & Motivation

This project will examine how the frequency of U.S. billion-dollar weather disasters has changed over time and how these changes relate to climate variables such as global temperature anomalies. This is an applied project using publicly available NOAA climate data.

Data

The primary dataset will be the NOAA Billion-Dollar Disasters annual series (1980–present), giving the number of billion-dollar climate and weather events per year.

Year (scaled as in class),

Annual disaster counts (response),

Temperature anomaly and related transformations (lags, squares, interactions).

Research Questions

What is the nonparametric relationship between time, temperature, and disaster frequency?

Is the increasing trend statistically significant, and how large is the uncertainty?

Can flexible models (neural nets) predict disaster counts effectively compared to linear smoothers?

When testing many possible predictors, which remain significant after controlling the False Discovery Rate?

Expected Deliverables

The final report will include:

Data description and preprocessing,

Smoother fits and PRESS comparison,

Bootstrap distributions and confidence intervals,

Neural network training results, parity plots,

FDR plots and interpretation,

Discussion of model performance, significance of trends, and limitations.

Why This Project Fits the Class

This project integrates nearly every major statistical tool taught in class—linear smoothers, PRESS, bootstrap, neural networks, and FDR—within one coherent applied analysis. It also uses real, non-toy data.